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OPTIMIZING PERFUSION CONDITIONS FOR 3D NEUROVASCULAR NETWORKS ON CHIP

Bachelor's thesis
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ABSTRACT

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The neurovascular system is a crucial functional entity that integrates the neural and vascular networks, which serve as the two primary transport systems within organisms' bodies, providing an interdependent structural and functional basis for each other. These networks cover the entire body, facilitating the transport of various physiochemical substances; the neural network transmits electrochemical signals, while the vascular network delivers oxygen, signaling molecules and nutrients to all tissues via blood flow. Therefore, neurovascular modeling enhanced by medium perfusion represents an essential development for *in vitro* approach to acquire scientific information about these innervated tissues and their complex cooperation. Perfusion flow is crucial for cell viability because dynamic flow affects many cellular functions, such as signaling and gene expression in tissues; without this flow, authentic tissue function cannot be generated in the laboratory. Neurons and vascular cells also share numerous functional molecules and collaboratively maintain whole-body homeostasis. It is crucial for *in vitro* research to devise new methods to make these components more realistically integrated in multiculture modeling, using dynamic perfusion techniques that mimic *in vivo* tissues. This thesis explores the use of a rocking platform to induce dynamic bidirectional perfusion in neurovascular multicultures. The principal aim was to see how perfusion with rocking platform affect the hydrogels and neurovascular cell growth compared to standard hydrostatic pressure gravity-driven perfusion.

This study investigated the effects of the perfusion mechanism on various cell cultures, all sharing the same hydrogel composition, chip and medium, but differing in cell type: neuron monoculture chips; vascular chips composed of bone marrow stromal cells (BMSCs) and human umbilical vein endothelial cells (HUVECs); and finally neurovascular chips which integrated neurons, BMSCs, and HUVECs. In addition to these, there were controls from all cell type cultures that were perfused with hydrostatic pressure.

According to scientific observations, the results indicated that the rocker platform perfusion did not yield a cell-stimulating effect compared to the control cultures, which were perfused by hydrostatic pressure gravity-driven perfusion. The hydrogels showed degradation and shrinkage, which was not seen in the controls i.e. gravity-driven perfusion, so the rocker plate parameters were not optimal for neurovascular cultures. Instead, neuron monocultures in both perfusions appeared to be spared the degradation of the hydrogels.

However, more studies are needed as there are very few studies of neurovascular perfusion outside of modeling the blood brain barrier. The information obtained from this in vitro modelling experiment can be applied in the future to support tissue modelling studies with neuronal and vascular networks, particularly with regard to perfusion.

Keywords: 3D cell culture, human derived cells, human induced pluripotent stem cells, hydrogel, microfluidic systems, neurovascular networks, organ-on-chip, vascularization

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TIIVISTELMÄ

Riku Korpiaho: Perfuusio olosuhteiden optimointi 3D-neurovaskulaariverkostoille kudossiruissa
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Neurovaskulaarinen järjestelmä on keskeinen toiminnallinen kokonaisuus, joka integroi hermo- ja vaskulaariverkostot toisiinsa, jotka toimivat kahtena pääosallisena kuljetusjärjestelmänä organismien kehoissa ja tarjoavat toisilleen toisistaan riippuvaisen rakenteellisen ja toiminnallisen perustan. Nämä verkostot kattavat koko ihmisen kehon, toimittaen erilaisia fysiokemiallisia substansseja; neuroverkot siirtävät sähkökemiallisia signaaleja, kun taas verisuonisto kuljettaa happea, signaalimolekyylejä sekä ravinteita kaikkiin kehon kudoksiin verenkierron avulla. Siksi mediumin eli elatusaineen läpivirtauksella tehostettu neurovaskulaarinen mallinnus on olennainen kehitys *in vitro* -soluviljelyyn, tieteellisen ymmärryksen hankkimiseksi näistä hermotetuista kudoksista ja niiden monimutkaisesta yhteistyöstä. Perfuusiovirtaus on ratkaisevan tärkeää solujen elinkelpoisuudelle, koska dynaaminen virtaus vaikuttaa moniin solutoimintoihin, kuten signalointiin ja geenien ilmentymiseen kudoksissa; Ilman tätä virtausta aitoa kudostoimintaa ei voida tuottaa laboratoriossa. Neuronit ja vaskulaarisolut jakavat monia oleellisia toiminnallisia molekyylejä sekä yhdessä ylläpitävät koko kehon homeostaasia. *In vitro* -tutkimuksessa on tärkeää kehittää uusia menetelmiä, joilla nämä komponentit integroidaan realistisemmin monikudosmallinnuksessa käyttämällä dynaamisia perfuusiotekniikoita, jotka jäljittelevät *in vivo* -kudoksien jatkuvaa virtausta. Tässä tutkielmassa keinutusalustan käyttöä testattiin dynaamisen kaksisuuntaisen perfuusion tuottamiseksi neurovaskulaarisissa kudossiruviljelmissä. Päätavoitteena oli nähdä, miten perfuusio keinualustalla vaikuttaa hydrogeeleihin ja neurovaskulaaristen solujen kasvuun verrattuna perinteiseen hydrostaattiseen painovoimaperfuusioon.

Tässä tutkimuksessa selvitettiin perfuusiomekanismin vaikutuksia eri soluviljelmiin, joilla kaikilla on sama hydrogeelikoostumus, siru ja väliaine, mutta jotka eroavat tyypeiltään: neuroniviljelmät joissa vain neuroneita; verisuoniviljelmät, jotka koostuvat luuytimen stroomasoluista (BMSC) ja ihmisen napalaskimoiden endoteelisoluista (HUVEC); ja lopuksi neurovaskulaariset viljelmät, jotka integroivat neuronit, BMSC:t ja HUVEC:t. Näiden lisäksi kaikista solutyyppiviljelmistä oli siis kontrollit, jotka olivat perfusoitu hydrostaattisella paineella.

Tieteellisen periaatteen mukaan tulokset osoittivat, että keinualustan perfuusio ei tuottanut soluja stimuloivaa vaikutusta verrattuna kontrolliviljelmiin, jotka perfuusioitiin hydrostaattisella painovoimaperfuusiolla. Keinutuslevyn viljelmien hydrogeeleissä havaittiin hajoamista ja kutistumista, jota ei havaittu kontrolleissa eli hydrostaattisessa painovoimaperfuusiossa, joten keinulevyn parametrit eivät olleet optimaalisia neurovaskulaarisille viljelmille. Sen sijaan neuronimonoviljelmät molemmissa perfuusioissa näyttivät säästyvän hydrogeelien hajoamiselta.

Lisää tutkimuksia kuitenkin tarvitaan tulosten todentamiseksi, ja neurovaskulaarisesta perfuusiosta on edelleen hyvin vähän tutkimuksia veri-aivoesteen mallintamisen ulkopuolella. Tästä *in vitro* -mallinnuskokeesta saatuja tietoja voidaan tulevaisuudessa soveltaa tukemaan kudosmallinnustutkimuksia, jotka kohdistuvat hermo- ja verisuoniverkoilla erityisesti perfuusion kontekstissa.

Avainsanat: 3D soluviljely, hydrogeeli, ihmisen indusoidut pluripotentit kantasolut, ihmisperäiset solut, mikrofluidiset järjestelmät, neurovaskulaariset verkot, organ-on-chip, vaskularisaatio

Tämän julkaisun alkuperäisyys on tarkastettu Turnitin Originality Check -ohjelmalla.

PREFACE

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Riku Korpiaho

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ABBREVIATIONS AND TERMS

2D	two-dimensional
3D	three-dimensional
BBB	blood brain barrier
BoC	body-on-chip microfluidic culture
CNS	central nervous system
ECM	extracellular matrix
EGM-2	Endothelial Cell Growth Medium-2
ECs	endothelial cells
EVT	extrasynaptic volume transmission, form of cell signaling
GFP	green fluorescent protein
HBMECs	human brain microvascular endothelial cells
HUVECs	human umbilical vein endothelial cells
in vitro	research technique outside the organism, artificial environment
in vivo	research technique inside the organism, in living tissue
mechano -transduction	cells physicochemical response to mechanical stimuli
NVU	neurovascular unit, a recurrent tissue unit
OoC	organ-on-chip microfluidic cell culture
organoid	simplified version of organ *in vitro*
paracrine	short distance cell signaling with chemical messengers
perfusion	throughput, through drainage, flow of fluid through the tissue
hiPSC	human induced pluripotent stem cells
iPSC	induced pluripotent stem cells
transwell culture	passive *in vivo* cell culture
VEGF	vascular endothelial growth factor
α-MEM	Minimum Essential Medium α

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1. INTRODUCTION

The central nervous system (CNS) and neurons in general, are of research interest for their still partly obscure ability to convert electrochemical impulses into subjective thoughts and vice versa, which is already a major scientific question. In addition, the CNS is also responsible for homeostasis in the body. The first theoretical outline of a nerve cell dates back to Emanuel Swedberg's studies of anatomy and physiology in the 1730s, after which in 1873 Camillo Golgi invented the "black reaction" to see the cell actually under a microscope, which led to the current neuron doctrine (Finger, 2004). Today, neurons and other cell types in brain tissue, such as glial cells, can be cultured outside the body due to Harrison's groundbreaking research in 1910, which laid the foundation for modern modelling based on *in vitro* cell cultures (Millet and Gillette, 2012).

A single neuron consists of soma, both exporter and importer branches (nerves) and their synapses. Neurons that do not belong to the CNS, i.e. the brain or spine, are peripheral neurons that transmit information further outside the CNS. Another important messaging system is the vascular network, transporting energy between tissues, but also participating in the transport of hormones, oxygen, and immune cells. At the CNS level, the interaction of these major systems is characterized by their common biology, such as common receptors and signaling molecules, as well as the effects of the systems on each other. Therefore, co-cultivation of these systems is important to achieve realistic tissue modeling. (Lok et al., 2007)

In organ-on-chip (OoC) culture models, medium nutrients, are applied by gravity onto the cells, with diffusion or through different types of circulation systems. When the aim is to make an actual model that mimics brain tissue, not only the relevant cells are needed, but also a suitable perfusion to improve vascularization. It has been noticed from measurements that even if the medium is switched to a gravity-based culture every day, it is still not enough to keep the flow state constant (Mykuliak et al., 2022; Winkelman et al., 2022). Different possibilities for more dynamic perfusion are now being sought, for example rocking plates, which keep the medium moving continuously.

Realistically perfused neurovascular cultures are essential for the development of OoC tissue modelling for research purposes for many reasons. Only 11% of drugs that have been found to work in animal studies are effective and safe for human use, which is main

reason to develop *in vivo* like tissue models using human derived cells for medical science. Another special feature of OoC modelling is that stem cell technology can be customized to each patient. (Miller and Shuler, 2016)

As part of this thesis, in addition to the literature section, an experimental part was carried out in which *in vitro* cell culture was tested in two different medium flow-through methods in practice. While the literature section covers an overview of tissue biology, cell culture and advances in artificial blood perfusion in recent years, the experimental section took rocker platform-induced perfusion as a practical example. In this study, cell cultures with perfusion induced by the rocking platform were compared with controls on standard gravitational-based perfusion performed at the same time. Three different culture types were used in both perfusion techniques, neuron, vascular and neurovascular. At the end of the thesis, the results are discussed and compared to the literature.

2. LITERATURE REVIEW

2.1 Neurovascular networks

The vascular network of vertebrates develops in a process called vasculogenesis, where, guided by local signals, angioblasts enter the emerging area of the tissue. A process called angiogenesis on the other hand, means branching already existing blood vessels. CNS is mostly vascularized by angiogenesis (Biswas et al., 2020). The brain of an adult human has a vascular network approximately 643 kilometers long when straightened, and throughout the body a total of about 5 liters of blood circulating at the same time (Kisler et al., 2017). The brain uses about 20 percent of the oxygen and energy in this blood volume, even though its weight is only 2 percent of the total body weight, underlining the importance of the brain as the body's overall "command center" and the importance of the vascular circulatory system to maintain its function (Raichle et al., 2001). In the immediate vicinity of each brain cell there is a blood vessel, no more than about 15 micrometers away (Tsai et al., 2009). These structures are inseparable in human biology but have been challenge to combine in tissue modeling.

The neurovascular unit (NVU) represents a complex multicellular entity that regulates blood circulation, transfers nutrients and metabolites in the brain tissue, on which the brain is completely dependent. The cell types present in the unit, such as neurons and endothelial cells in natural tissue, along with glial cells and extracellular matrix. In addition to these, mural cells, including smooth muscle cells and pericytes are also part of the unit, crucial for blood vessel development. The purpose of endothelial cells (EC) is to act as a "selective membrane" in the veins, allowing only selected molecules into the surrounding tissues. It should be mentioned that astrocytes are the most abundant glial cells, responsible for the strength of synapses, neurotransmitter uptake, and GABA release, but also regulate blood flow through their end feet attachments to veins. (Lenk et al., 2016; Schaeffer and Iadecola, 2021)

In vitro NVU (Figure 1) and neurovascular modelling in general have mainly focused on modelling the blood-brain barrier (BBB), which is essential in the study of drug transport and permeability. The endothelial cells of the BBB form a barrier that prevents the transfer of neurotoxic plasma components, xenobiotics and pathogens to the brain. (Jagadeesan et al., 2020) The crucial point to comprehend at this BBB-on-chip context is that the

flow traversing through the lumens of blood vessels and neurons are distinct. Therefore, in many studies, chip partitioning into two distinct channels is commonly observed, where rapid flow and shear stress prevail in the vascular channel, up to 50 ml/100g/min (Amirifar et al., 2022; Brown et al., 2015). When examining neurovascular tissue modeling outside the BBB, there are notably fewer published research studies.

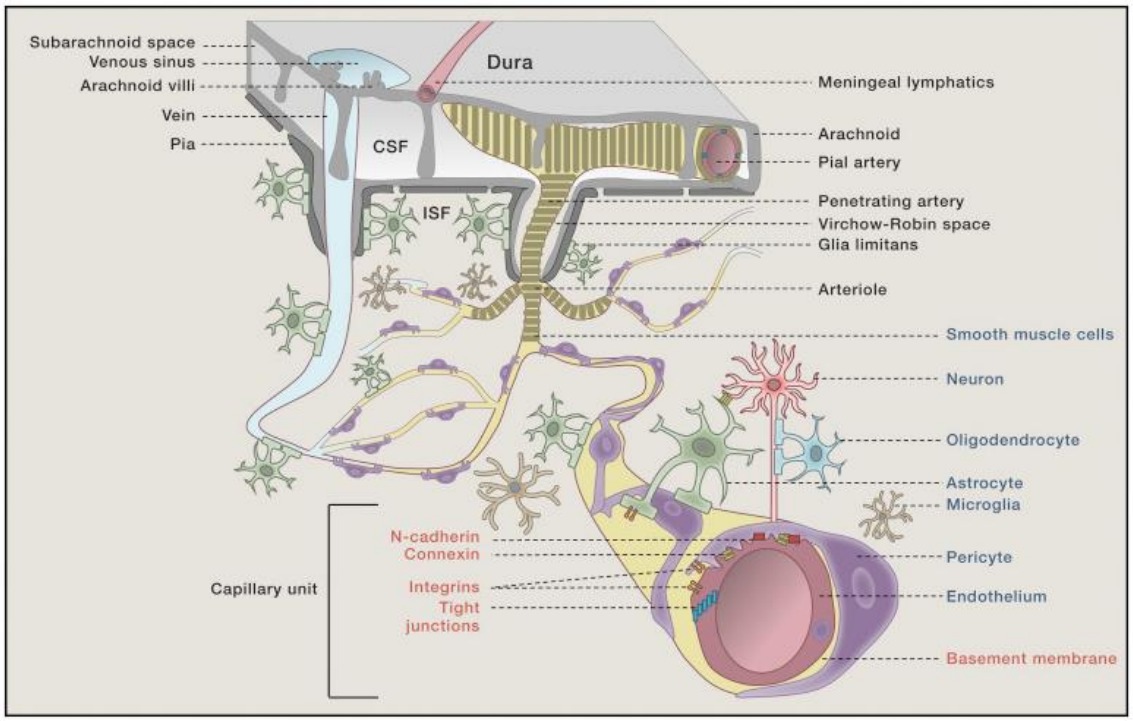


Figure 1. *Neurovascular unit (NVU). (Zhao et al., 2015).*

Neurovascular multiculture research has begun to progress out of monoculture neuronal or vascular models for a variety of reasons. For example, when studying diseases, disrupted coupling and signaling play a key role in their development. Without all cell types involved in signaling, a credible model cannot be built. The nervous system and vascular networks share countless signaling and control molecules with each other, of which vascular endothelial growth factor (VEGF) can be cited as the most studied, as well as endothelins, slits, netrins and semaphorins (Lok et al., 2007). For example, VEGF is strongly present in the formation of new veins but also in the proliferation of neurons, because it participates in crosstalk between these cells. Nerve cells strongly promote angiogenesis because the neurons themselves secrete factors into their environment. Based on these, it is clear that these systems strongly share a common evolutionary origin. (Isosaari et al., 2023; James and Mukouyama, 2011; Uwamori et al., 2017) Multicultures, i.e. neurovascular models in this case, are therefore an undeniable way to understand the communication, cooperation of neurons and vascular cells from the point of view of tissue modelling, developmental biology and drug research.

Technically, in stem cell-based *in vitro* cultures most commonly today as protocols evolve, neurons differentiated from human induced pluripotent stem cells (hiPSC) are used, with patient-specific medicine in mind, bringing the patient's genotype into the culture (Jagadeesan et al., 2020). That implies as well that hiPSC culture can be disease specific (Palma-Florez et al., 2024). iPSC stem cells have been created through a method discovered as a breakthrough in stem cell research, by restoring a somatic cell back to a stem cell using a special molecular cocktail (Takahashi and Yamanaka, 2006). The EC cells needed for the angiogenesis of cultures *in vitro* can be differentiated from hiPSCs like neurons, but it is still very difficult to mimic the complex organization of natural microenvironmental cues (Sun et al., 2022). Mural cell differentiation from hiPSCs is already possible with a clear protocol (Yamamoto et al., 2020). Ideally, all required cell types could be differentiated directly from hiPSCs.

2.2 3D Organ-on-chip microfluidic systems

Non-perfused, static 2D cell culture systems such as transwell cell culture fail in that physiological shear stress is absent (Campisi et al., 2018). All cells inside the body on the other hand, experience shear stress, but also compression, hydrostatic pressure, interstitial fluid flow and contraction that affect the essence of the cells (Jagadeesan et al., 2020; Wang et al., 2023). Microfluidic chip cultures on the other hand, are characterized not only by their dimension and perfusion potential, but also by their potential interlinkability. This opens up diverse future prospects for cell culture research, such as the body-on-chip (BoC) concept, where cultures consisting of different cells in the body can be linked to each other, achieving organ-organ interactions. (Miller and Shuler, 2016)

Steps towards more sophisticated *in vitro* cultures are 3D OoC microfluidic systems, which generally increasingly aim at *in vivo* type tissue modelling in miniature. 3D microfluidic chips currently use high-biocompatibility, gas-permeable PDMS (polydimethylsiloxane), which has been the material of choice for culture substrates for a long time. The typical method for establishing 3D cultures involves incorporating cells into hydrogels, such as fibrin and collagen 1 hydrogel. The advantage of hydrogels in general is that they can mimic the extracellular matrix. (Lee et al., 2017)

Popular 3D hydrogel chip design from AIM biotech allows to generate perfusion by unidirectional hydrostatic pressure by putting different amounts of medium in the opposite chambers and allowing it to equilibrate. Interstitial flow passes through the hydrogel cell channel, as shown in Figure 2. This chip has become something of a standard, which is particularly good because results can be better compared between laboratories. (Winkelman et al., 2022) Whether this type of chip is the final solution or not is a different

question. However, analogous chip architectures have been postulated for the establishment of a functional microfluidic chip (mostly for BBB), wherein cellular positioning is achieved through the utilization of micropoles (AIM biotech), or a flow-guiding design (PhaseGuide technology, Mimetas). (Lyu et al., 2021)

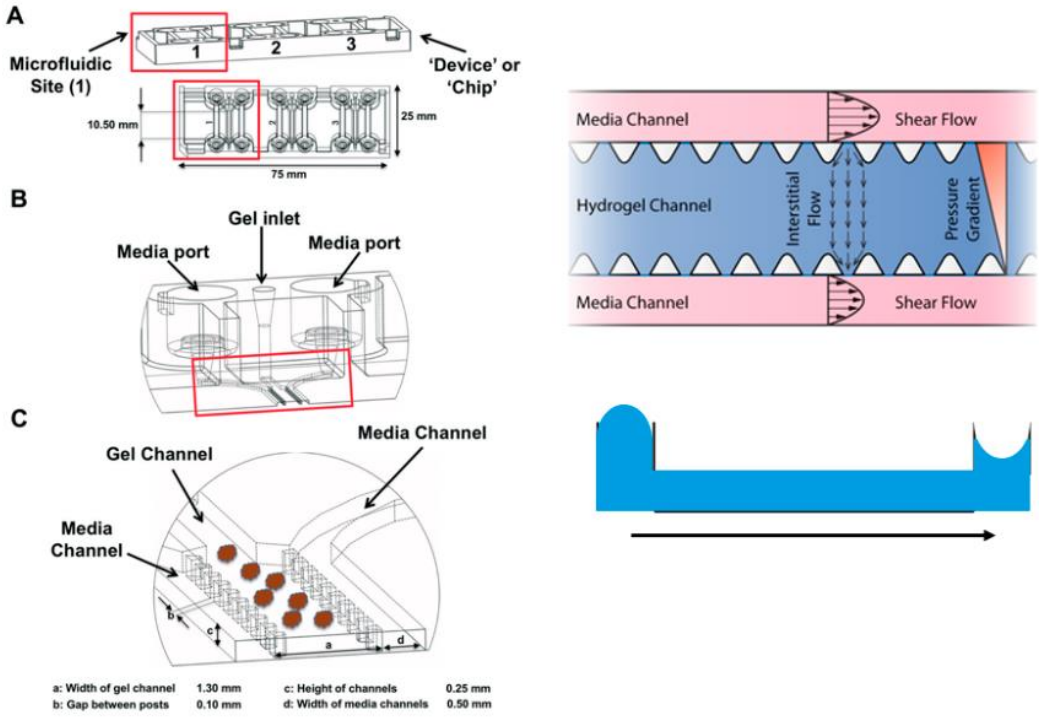


Figure 2. *AIM Biotech IndenTx chip design. (https://aimbiotech.com/).*

One of the undoubtedly most desirable goals of OoC models and more advanced BoC microfluidic systems is also to radically restrict animal testing, both for ethical reasons and because animals are simply not the same as humans as models. Hence, animal testing not provide a comprehensive picture of how drugs and diseases work in humans (Boos et al., 2019). The advantages of 3D microfluidic OoC models include tissue organization, dynamic properties of the system, biochemical function, structural configuration and perfusion. Undoubtedly, more realistic tissue models are expected to give a clear boost to drug development and lowering its costs from the current one. Human iPSCs are an intriguing prospect for personalized medicine and almost any type of differentiated tissue can be cultured on a microfluidic chip. However, a lot of optimizations is needed so that the culture medium, perfusion, differentiation protocol, cell line, hydrogel composition, chip structure, materials are suitable for all multicultural cell types. Much remains to be optimized in the context of neurovascular tissue *in vitro* OoC modelling.

2.3 Characteristics of perfusion flow

The normal cerebral blood flow (CBF) of the human brain from the literature is about 50 ml/100 g of tissue per minute through veins and arteries. However, *in vivo* there is variation between, for example, white matter and gray matter, of which the rate for gray matter is significantly higher. This assumes that the average human brain volume is 1.34 l and the flow rate is 0.69 l/min. (Madsen et al., 1993; Miller and Shuler, 2016; Price et al., 2003) In contrast to this, interstitial flow in the brain is relatively slow, varying from 0.1 to 0.3 µl/min in rats. (Abbott, 2004)

In cell cultures, the flow rate becomes a more complicated because there are countless types of perfusion systems and chip designs. 2D cell cultures, medium administration is commonly done in static method, gravitationally such as transwell culture but what has been noted before, they are completely devoid of shear stress and therefore do not correspond to *in vivo* tissue. Instead of transwell cultures, AIM BioTech (Figure 2) styled microfluidic chips can be used to create hydrostatic pressure flow through the 3D hydrogel channel.

In the study by Mykuliak et al. they find out when using gravity-driven perfusion on the fourth day of culture compared to the first, the flow rate drops from 3.1 µl/min to 0.35 µl/min but the flow duration increases from 1 hour to 5 hours (Mykuliak et al., 2022). Thus, gravity based hydrostatic pressure alone is not the most optimal method in the context of perfusion, as it does not remain stable. This has also been observed in other studies, with a neurovascular study by Winkelman et al., it was found that under hydrostatic pressure, perfusion ceases to resemble dynamic culture for about 15 hours per day after the pressure differences between the channels of the AIM chip (Figure 2) have equalized (Winkelman et al., 2022). Of course, this can be partially circumvented, as in Osaki et al., where large medium stocks coupled to a chip to maintain a steady hydrostatic pressure, with only a 15% decrease observed over a 24-hour period (Osaki et al., 2018).

Dynamic *in vitro* microfluidic systems have operated also with pumps, which involves problems such as dead volume, complex handling and connections. A pneumatic pump would be one option, but also requires complex external equipment, as one of the goals can be that the OoC system does not require complex equipment. In addition to this, pumpless designs are less expensive and they use low reagent volumes. (Lee et al., 2019) However, even in early 3D neuron cultures, it has been found that the optimal flow would be about 11 µl/min, which could correspond to up to 6-27 manual medium changes per day. (Cullen et al., 2006)

The rocking plate has also been used to cause continuous flow. In the Lyu et al study, a successful NVU-on-chip culture was performed on a rocking plate, where the rotation on the plate is between 7 degrees and – 7 degrees, with a bidirectional cycle of 16 minutes in which, however, does not mention the actual flow rate as a numerical value (Lyu et al., 2021). In such a system, the flow can be also unidirectional in several ways, it can be achieved by rotating the culture (rOoC) or on a rocking platform with complex design such as UniChip (Busek et al., 2022; Wang and Shuler, 2018). In addition to these OoC cultures, it is worth mentioning that as early as 2016, a functional 14-part modular BoC system perfused with a rocking platform was developed, characterized by a successfully implemented bidirectional flow and overall cell viability exceeding 85% (see Figure 3) (Miller and Shuler, 2016).

Organs	Calculated channel fluid volume (μL)	Calculated chamber fluid volume (μL)	Pipetted PGMatrix volume (μL)	Calculated linear flow rate (mm/s)	Observed linear flow rate (mm/s)	Calculated flow rate (μL/s)	Observed flow rate (μL/s)
Lung		5.1	25.3	5.90		1.02	
Skin	3.0	19.4	97	1.85	1.81 ± 0.56 (ave of 2)	0.10	0.10 (ave of 2)
GI tract	2.99	6.2	30.8	3.64	3.57 ± 0.90	0.39	0.38
Fat	2.34	17.9	89.7	0.36		0.03	
Kidney	3.36	1.6	8	4.08	4.46 ± 1.69	0.49	0.54
Heart	2.65	1.8	9	3.16	5.95 ± 0.87	0.30	0.57[a]
Adrenal glands	0.33	0.5	2.5	0.83	1.21 ± 0.44	0.01	0.01
Liver	3.56	7.9	39.3	4.33	4.1 ± 0.90	0.55	0.52
Spleen	1.46	1.1	5.5	1.73	3.27 ± 0.17	0.09	0.17[a]
Pancreas	0.76	0.5	2.5	1.11	2.11 ± 0.87	0.03	0.06[a]
Bone marrow	2.15	8.0	39.8	2.60	3.51 ± 1.00	0.20	0.27[a]
Brain	2.56	6.7	33.5	3.08	2.69 ± 0.40	0.28	0.24
Muscle	3.02	20	100	0.46		0.05	
Reservoirs		4,610	627.6 + 700 = 1,327.6				
Total volumes		Fluid volume (μL) 4,734.9	Matrix volume (μL) 1,810.5				

Figure 3. *Miller and Shuler 14-part modular body-on-chip system data, which incorporates a rocking platform to induce perfusion flow rates for organs. (Miller and Shuler, 2016).*

Flow rate plays a role in many variables. In addition to transporting oxygen and nutrients, the speed of flow extensively affects mechanotransduction in cells, which covers the cell's biochemical and electrochemical responses when they experience mechanical stimuli. The reception of perfusion flow rate in cells depends in turn, on the age of the culture and the method of cultivation. Certainly, various tissues within different organs are expected to exhibit distinct values, such as flow rates, reflecting their unique physiological characteristics (Table 1).

The phenomenon of extrasynaptic volume transmission (EVT) which entails short distance signaling between cells through the extracellular matrix with the assistance of flowing signal molecules, is overshadowed in neuroscience by synaptic communication. It is another method of communication between cells that must be considered especially when optimizing perfusion. Problems with culture may occur if perfusion is too fast because it washes out soluble signal molecules, but more research is needed on this. Too

fast flow has also been observed to confuse the synchronous spike bursts of neurons. The problem is also if the perfusion is too slow, because the shear stress does not affect as it should, and the cells do not develop realistically. (Herzog et al., 2021) Shear stress should be very close to the value in the microvasculature vessels of the brain which is 0.01-10 dyne/cm^2 (Lyu et al., 2021).

EVT can be divided into paracrine and long-distance signaling. Paracrine communication has been found to be particularly present with GABA and glutamate, as well as neurosteroids, endocannabinoids and nitric oxide. Long-distance extrasynaptic communication touches neurotransmitters such as dopamine and acetylcholine. Thus, EVT affects locally, for example, to electrical activity and junction strengths in cells. Especially in culture models, because synthetic nutrient solutions are used instead of blood, brain cells are known to be increasingly dependent on local secretion. (Herzog et al., 2021)

Perfusion flow also directly influences genetic pathways and endothelial vascular transformation. In a study conducted by Vatine et al. on human brain microvascular endothelial cells, it was found that the majority of the top 10 upregulated genes under a perfusion of 2.4 dyne/cm^2 shear stress compared to 0.01 dyne/cm^2. These findings thus suggested that upregulations occurred primarily through mechanotransduction mechanisms rather than other factors. (DeStefano et al., 2017; Vatine et al., 2019)

To summarize the phenomenon of shear stress, it can be said that although there are no direct *in vivo* studies in brain capillaries, it can be estimated to be more than 6 dyne/cm^2 (Wevers et al., 2021). Another thing about shear stress averaging is that it differs strongly depending on the thickness of the vasculature and the degree of curvature – but in general it can be said to vary widely depending on the vessel from 0.1 to 60 dyne/cm^2. For comparison, the Interstitial flow study by Winkelman et al. which utilized the hydrostatic pressure method produced 0.394 dyne/cm^2 shear stress on AIM Biotech chip. (Winkelman et al., 2022) In arteries, shear stress can be high as 80 dyne/cm^2. (Lee et al., 2019)

Table 1. *A collection of different flow characteristics and their relations.*

Source	Dimension	Shear stress	Chip type	Cell type	Flow characteristics
(Lyu et al., 2021)	3D	3.4 dyne/cm^2	"AIM biotech analog", experimental chip	Human primary BMEC, pericytes, astrocytes, microglial	Rocking platform, 4° degrees, tilt time 0.5 min, bidirectional flow
(Wang and Shuler, 2018)	2D	5.3 dyne/cm^2	UniChip	HUVEC	15 s interval, 18° degrees, unidirectional flow
(Lee et al., 2019)	3D	9.79 x 10^2 dyne/cm^2	"UniChip analog" with bypass channel	HUVEC	Rocking platform, Cycle: 20° tilting angle with 5-min dwell time, tilting speed 5°/s to −10° tilting angle, 0-min dwell time, speed 5/s to back 20°. Flow rate of 6.5 µL/min (0.4 µl/min per channel).
(Wevers et al., 2021)	3D	1.7 dyne/cm^2	Three-lane Organo-Plate	Immortalized, BMECs (TY10 cell line), Human brain pericytes (hBPCT cell line), human astrocytes (hAst cell line)	Rocking platform, 7° degrees, tilt time 8 min, bidirectional flow
(Pediaditakis et al., 2022)	3D	-	two-channel microengineered chip	primary astrocytes, pericytes, and iPSC-derived brain microvascular endothelial-like cells, human iPSC-derived cortical neurons and microglial cell line	Peristaltic pump, perfusion flow 60 mL/h, unidirectional flow
(Winkelman et al., 2022)	3D	0.394 dyne/cm^2	AIM BioTech chip	Primary human brain endothelial cells, pericytes and astrocytes	hydrostatic pressure gravity driven flow, unidirectional
(Jacob et al., 2016)	3D	0.01-10 dyne/cm^2	Natural tissue in the brain	All types of brain cells	Unidirectional, typical velocity in capillaries around 0.5 mm/s

2.4 Aims of the study

This experimental thesis aims to study different perfusion conditions used with 3D neurovascular networks on-chip model. Multiculture contains different cell types, hiPSC derived cortical neurons, green fluorescent protein (GFP)-tagged human umbilical vein endothelial cells (HUVECs) and bone-marrow derived stem/stromal cells (BMSCs) as a mural cell type origin. The microfluidic flow is obtained either by rocking platform induced bidirectional flow or traditional gravitational perfusion. Thus, the aim is to determine whether bidirectional flow is beneficial for the formation of 3D vascular, neurovascular, or neuronal networks. Therefore, the potential or non-benefits compared to gravitational perfusion can be considered. The information gathered from this experimental thesis will be applied to the design of future multiculture perfusion conditions.

The formulation of the hypothesis for this experimental study is premised on the evidence and findings from existing studies and literature, which have clarified the beneficial impact of perfusion on cellular growth especially in HUVEC cells. Based on this foundation, it is hypothesized that, within the context of this study, cells might exhibit enhanced growth efficiency on a rocker platform as compared to a traditional culture environment. This anticipated outcome is attributed to the modulation of hydrostatic pressure, suggesting that the dynamic conditions provided by the rocker platform could foster more favorable conditions for cell proliferation, although this possibility depends on a lot of variables.

3. MATERIALS AND METHODS

The experimental part of the bachelor's thesis was completed together with student Siiri Sihvonen.

3.1 Cells

As in a previously published study, 3 different cell types were used to establish multicultures, HUVECs, hiPSC-derived neurons, and isolated BMSCs. (Isosaari et al., 2023).

Pooled HUVECs which was already modified to express green fluorescent protein (GFP) by Angio-proteomie, were used at passage number 5. Prior to chip experiments, the HUVECs were expanded in Endothelial Cell Growth Medium-2 (EGM-2, Lonza) with 2% human serum (Serana) (Isosaari et al., 2023).

For neuronal differentiation, the in house derived hiPSC cell line UTA.04511.WTs was used. Furthermore, the cell line was derived at the MET Faculty of Medicine and Health Technology, iPS Cells facility, Tampere University, Finland. The differentiation and expansion from stem cells into cortical neurons were implemented following a previously published in-house protocol by laboratory technicians. The differentiation included induction, proliferation and maturation phase of the cells (Hyvärinen et al., 2019). The project has received ethical approval from the Ethics Committee of the Expert Responsibility area at Tampere University Hospital to use the specified hPSC line in neuronal research (R20159), ensuring that all ethical considerations are adequately addressed.

BMSCs, stem cells/stromal cells derived from human bone marrow, were isolated during surgery from bone marrow samples from donors at the Department of Orthopedics and Traumatology, Tampere University Hospital (Wang et al., 2019). Initial medium for BMSCs was α-MEM, (Gibco) + 5% human serum (Serana) + 1% penicillin/streptomycin (Lonza) + 5 ng/ml basic fibroblast growth factor (FGF-B, Miltenyi Biotec). Samples were obtained with each donor’s written informed consent and processed with the ethical approval of the Ethics Committee of the Expert Responsibility area of Tampere University Hospital, Tampere, Finland (R15174).

Before HUVECs and BMSC cells were included in the experiments, student Susanna Harju was responsible for their thawing and maintenance until plating.

3.2 Cell culture platform

The microfluidic chips used for the perfusion optimization experiment are 3D cell culture chips from AIM Biotech (3D Cell Culture Chip, AIM Biotech, Singapore). Each chip features a 3-channel design, indicating the capability to conduct three simultaneous cultivations. A schematic picture of the chip can be seen before in Figure 2. These chips are made of PDMS, that is permeable to gases (Lee et al., 2017).

Each of the three hydrogel channels is bordered by two media channels. The addition of varying volumes of medium to the upper and lower media channels generates interstitial flow of medium through the hydrogel channel, driven by the hydrostatic pressure differentials established within the system. In the experiment, the perfusion flow within the rocker cultures occurs through the rocking motion of the platform rather than being driven by quantitative differences in fluid volumes as seen in Figure 4.

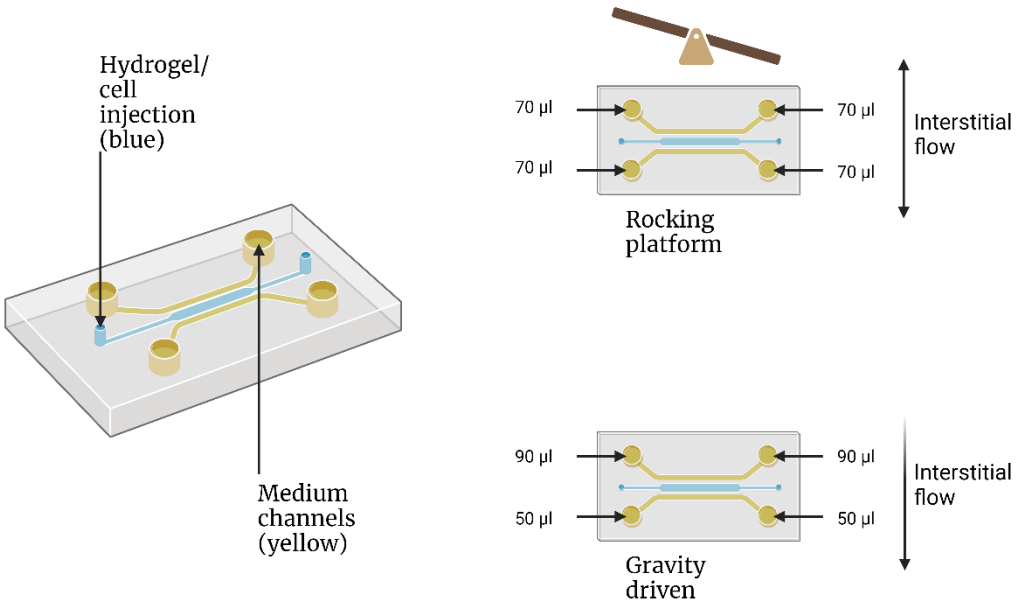


Figure 4. *A schematic image of the media volumes and hydrogel added to the AIM Biotech chip channels, noting that there are three of these scrolling on one chip. Created with Bio-Render.com.*

3.3 3D neurovascular networks on chip cultures

Cells were plated to the chips according to the previously published protocol. (Isosaari et al., 2023)

The hydrogel was mixture of 1mg/ml rat tail collagen (Gibco), 1.5mg/ml human fibrinogen (Lonza) and 0.6iU/ml thrombin (Sigma-Aldrich). The hydrogel solution was prepared just before plating, into which the desired cells were added in the right proportions depending on chip (neurovascular, neuro, vascular) type.

Neurons were plated at density 2.5 x 10^6 cells per ml, where HUVECs at 5 x 10^6 cells/ml density and at last, BMSCs were incorporated at 1 x 10^6 cells/ml. This entity was injected with a pipette into the central gel channel into AIM Biotech chip, which is illustrated in Figure 4. As is obvious, only neurons were injected into the neurochips, but BMSCs and HUVECs were added to the vascular chips together, while all the above cell types were injected into the neurovascular chips.

10 µl of cell-laden hydrogel was seeded every gel channel of every chip. In this study we had 12 where 6 chips for neurovascular design, 4 chips for neurons only and 4 chips for vascular cultures. After injection, the chips were turned over for 30 seconds to ensure even distribution of cells in the Z-direction, after which they were transferred to the incubator for 30-40 minutes.

After incubation, first mediums were added for all chips as shown in Figure 4. Initially, 15 µl of medium was added to all medium channels, after which the reservoirs were filled with either 70/70 µl for rocker perfusion chips or 90/50 µl for gravity perfusion chips. The medium utilized in the study for all chips was EGM-2 (Endothelial Cell Growth Medium-2), supplement with aprotinin (Abcam) at a final concentration of 40 µg/ml to prevent gel degradation.

Rocker platform (Mimetas OrganoFlow) specifications were set to 10 min rocking interval, where the angle oscillates between 7° and -7° degrees, as described in the triculture study by Wevers et al. (Wevers et al., 2021). To maintain humidity, a small amount of water was placed onto a reservoir on the plates where the microfluidic chips were cultured.

Medium was refreshed to rocker chips three times a week, Monday, Wednesday, and Friday on a cycle. For all other chips that were not on rocker plate, i.e. gravity-based controls, the medium was changed every day, except for one test chip which had same medium refreshment interval as the rocker chips. The purpose of this one test chip was

to illustrate whether a neurovascular culture with gravity perfusion could survive with a longer exchange interval.

3.4 Live imaging

Our major imaging instrument was Leica Dmi8 inverted fluorescence microscope (Leica) with which all images acquired. Each image was taken in such a way that a phase contrast image and a fluorescence image were taken at the same time. The cultures were imaged on days 2,5,7,9,12, and 13. Image editing such as contrast leveling was done with Leica's own program LAS X.

4. RESULTS

In this thesis, the aim was to compare more traditional gravity perfusion with another kind of perfusion on a rocker platform in multiculture modeling. Object of interest was mainly neurovascular cultures on a rocking platform perfusion compared to hydrostatic pressure perfusion. At the same time, in addition to neurovascular chips, neuron monocultures and vascular chips (ECs and mural cells) chips with both perfusion methods were also taken for comparison.

4.1 Hydrogel integrity

The hydrogels containing vasculature forming cells on the rocker platform chips were visually observed to begin degrading on the day 5 as seen in Figure 5. They appeared to shrunk, which was not seen in any of controls in hydrostatic pressure cultures throughout the 14-day culturing period. In addition to the regression of the hydrogel, vascular cell growth was clearly lower in the cultures on the rocker platform overall.

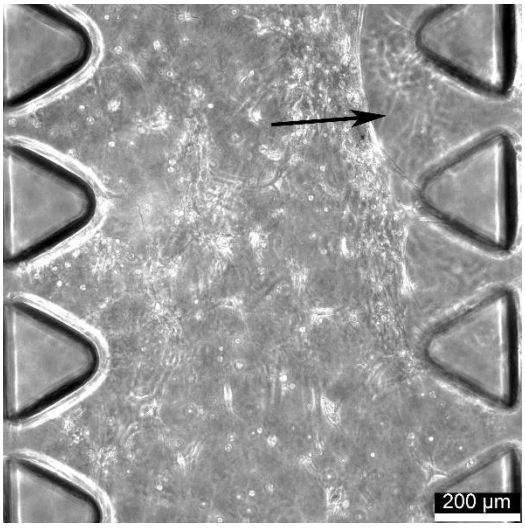


Figure 5. *A typical hydrogel shrinkage observed at day 5 on vascular rocker chip phase contrast photo. Black arrow shows shrinkage. Scale bar: 200 µm.*

4.2 Neuron monocultures with both perfusion methods

Neurons grew very typically on both perfusion conditions. Long neurites can be observed in their final days as shown in Figure 6. Medium what was used in this study was not most optimal for neurons, but they still proliferate on EGM-2 vividly. Neuronal

monocultures had preserved their hydrogel well in rocker platform perfusion, which was not seen in the presence of vascular cells (neurovascular and vascular cultures).

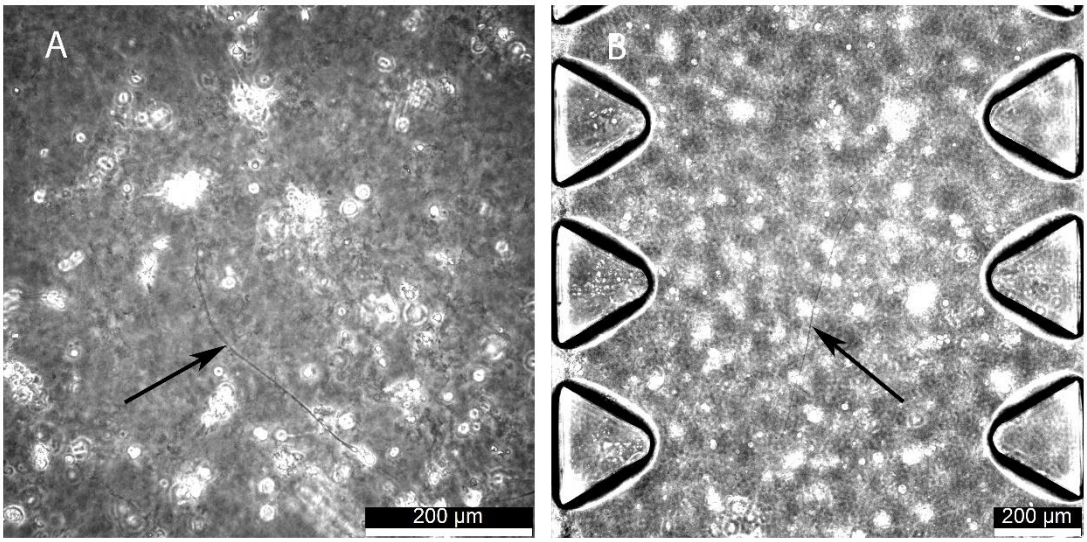


Figure 6. *A neuron axon (black arrow) on both perfusions. Picture A shows a rocking platform culture at day 12, and B represents a gravity-driven culture at day 13. Hydrogels remained in better condition in neuron monocultures also in rocker perfusion. Scale bar: 200 µm.*

4.3 Rocker platform perfused chips

It was evident that the HUVEC cells remained mostly round and hardly developed vasculature on rocker platform, throughout the culturing period of 14 days. There was a total of 2 neurovascular rocker platform chips, both with 3 separate cultures. Some signs of the beginning of vascularization can be seen on some samples but overall, it can be stated that vascularization is not present clearly in neurovascular rocker chips, which can be seen in Figures 7-8, where the same chip is shown in both figures. In Figure 9, another rocker platform neurovascular chip culture is shown where more developed vascularization is visible.

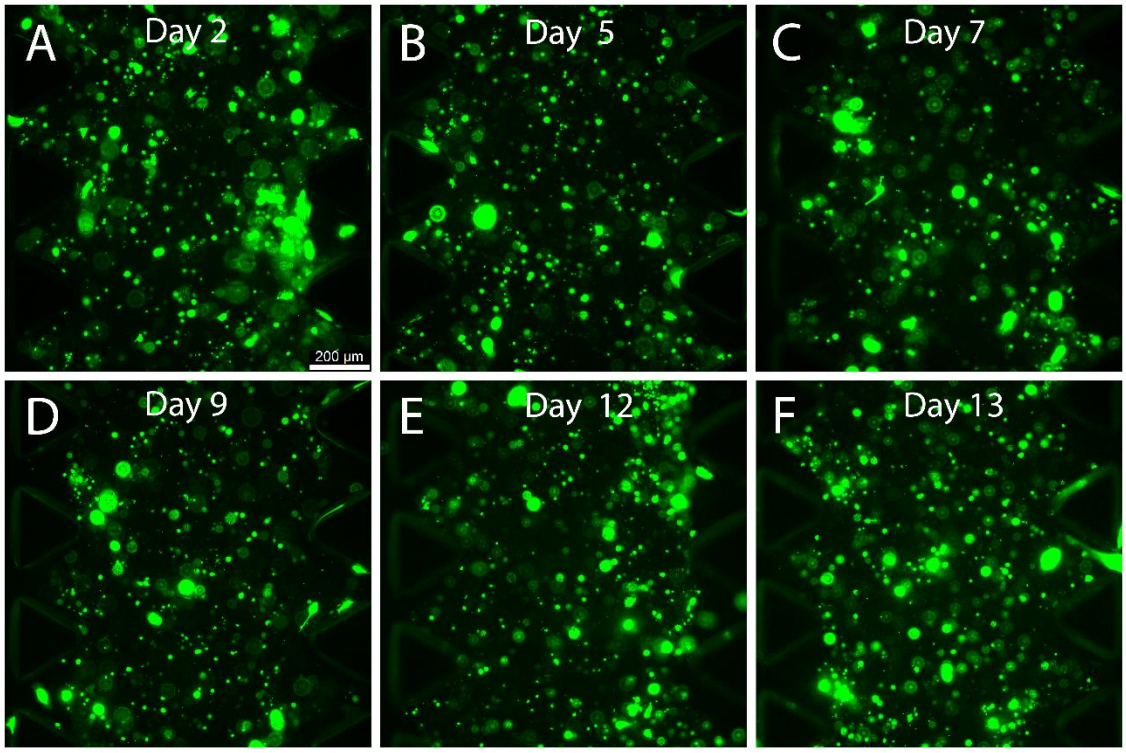


Figure 7. *Neurovascular culture with rocker perfusion, all 6 timepoints are showing no substantial vascularization. Scale bar: 200 µm.*

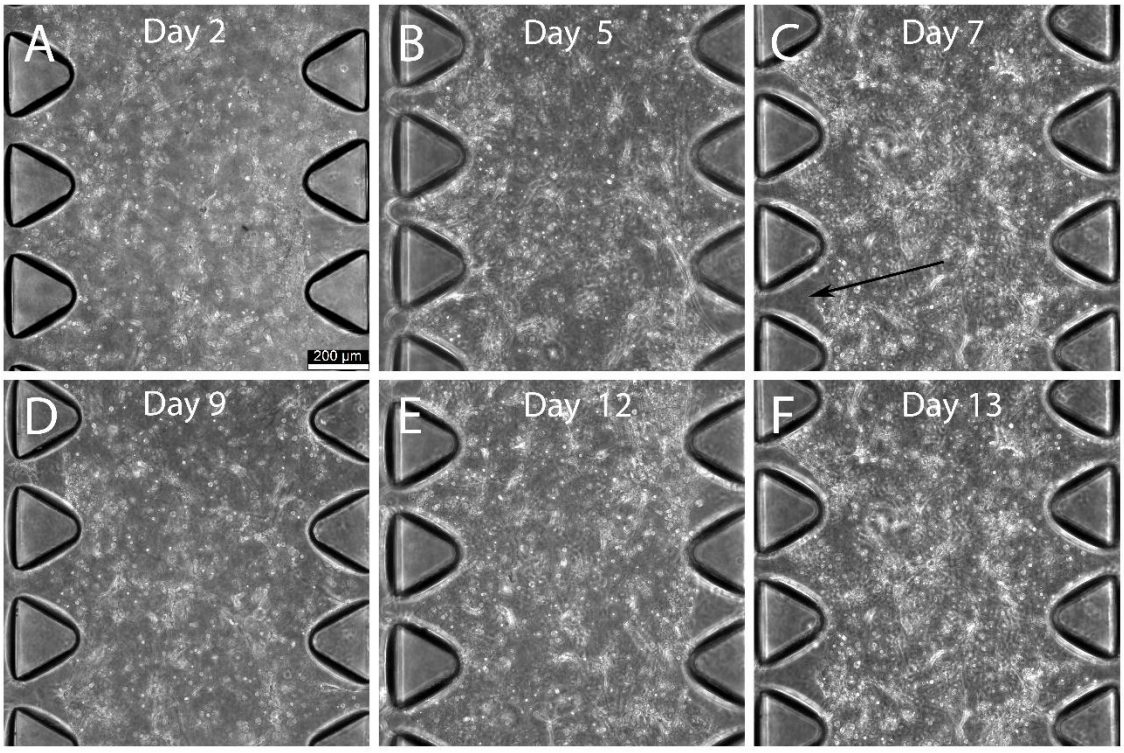


Figure 8. *Neurovascular rocker culture in phase contrast images. Black arrow shows shrinkage. Scale bar: 200 µm.*

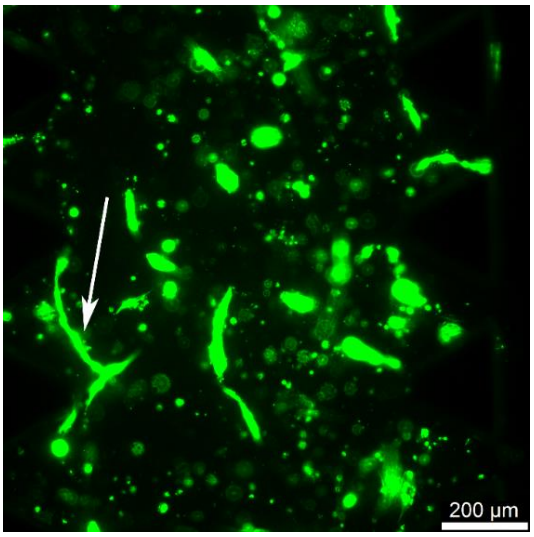


Figure 9. *Some vascularization were detected on the neurovascular cultures on rocking platform at day 9 in different samples than the culture shown in Figures 7-8, indicating variation in vasculature formation inside the sample group. White arrow points to vascularization. Scale bar: 200 µm.*

Despite the lack of vasculature formation on neurovascular chips shown above, some visually thick vascularization were detected in pure vascular rocker chip in combination with hydrogel shrinkage shown in Figure 10. Figure 11 shows the same image area and chip as phase contrast images showing the degradation of the hydrogel.

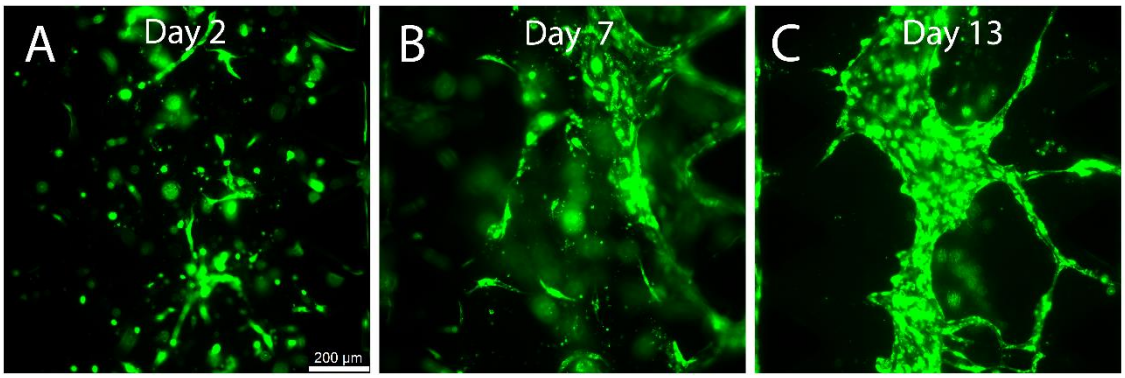


Figure 10. *Vascular rocker chip with noticeable vascularization. Scale bar: 200 µm.*

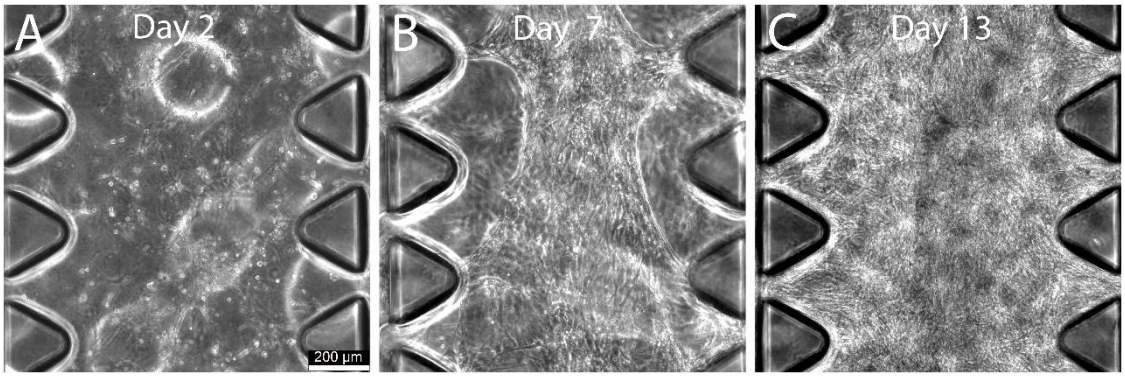


Figure 11. *Vascular chip on rocker perfusion in phase contrast images, where substantial degradation can be seen in picture B. In picture C, the sample appears improved; however, it was taken from a different section of the culture to demonstrate that the gel was partially in better condition. Scale bar: 200 µm.*

4.4 Gravity-driven hydrostatic pressure controls

The interpretability of the results is complicated by the weakness of neurovascular control growth in chips perfused by hydrostatic pressure, as seen in Figures 13-14. In any case, clearly a more developed vascular network was detected with gravity-driven perfusion than what was seen on the respective rocker platform cultures.

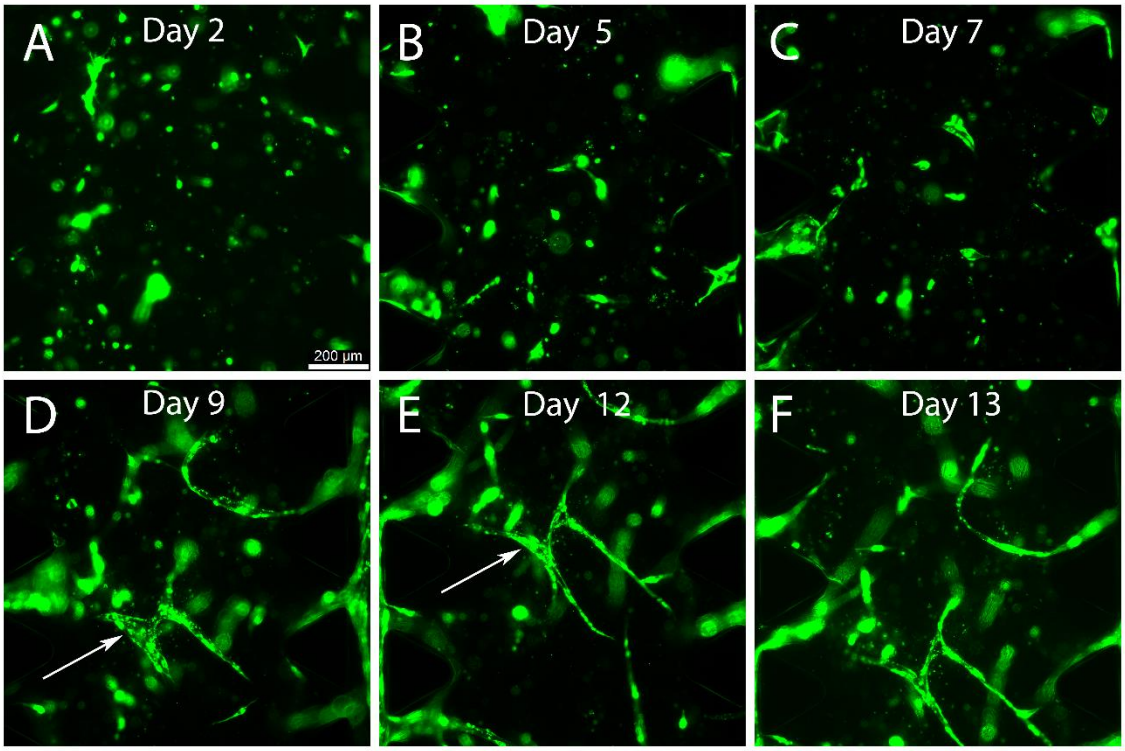


Figure 12. *Neurovascular control culture with gravity-driven hydrostatic pressure perfusion. White arrow indicates vascularization. Scale bar: 200 µm.*

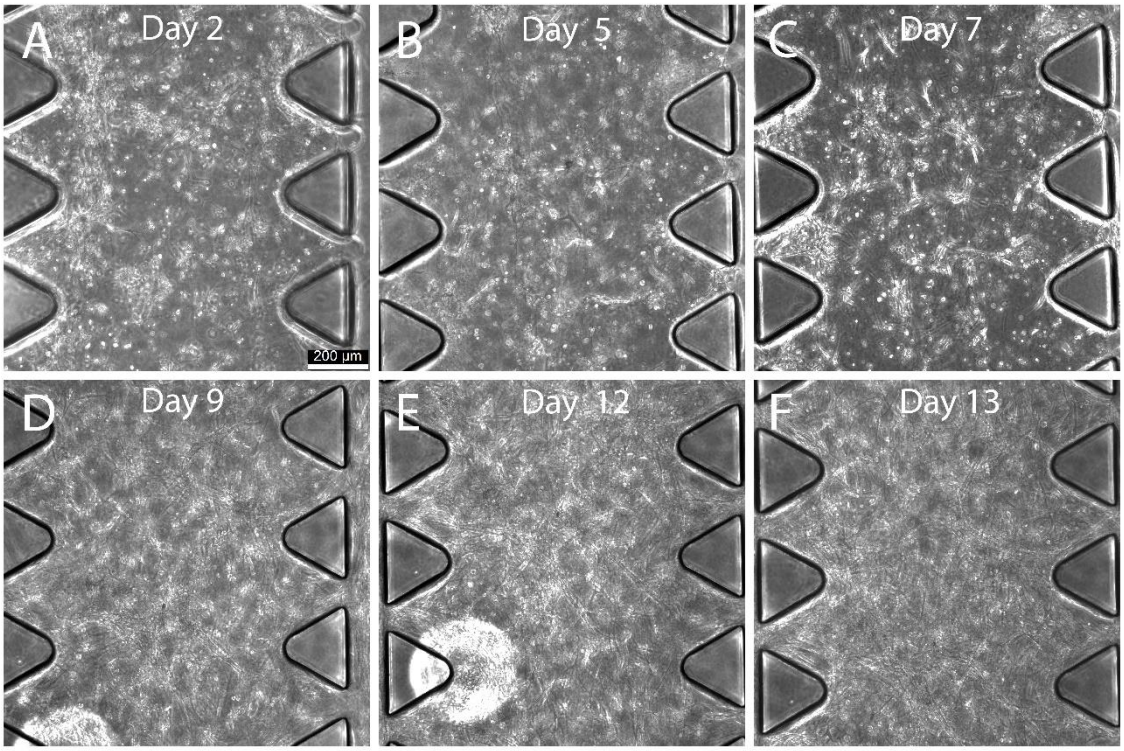


Figure 13. *Neurovascular control culture with hydrostatic pressure driven perfusion on phase contrast picture. No degradation of the hydrogel was observed. Scale bar: 200 µm.*

Interestingly, the neurovascular cultures that received medium refreshment only three times a week were more developed than the rocker cultures, which can be seen on Figure 14. Nevertheless, the development is still weaker in the chips with a refreshment period of 3/7 days than in the chips where the culture medium was refreshed every day. According to the standard protocol, in all hydrostatic gravity driven cultures, the medium is changed every day, as was done in this study with the controls.

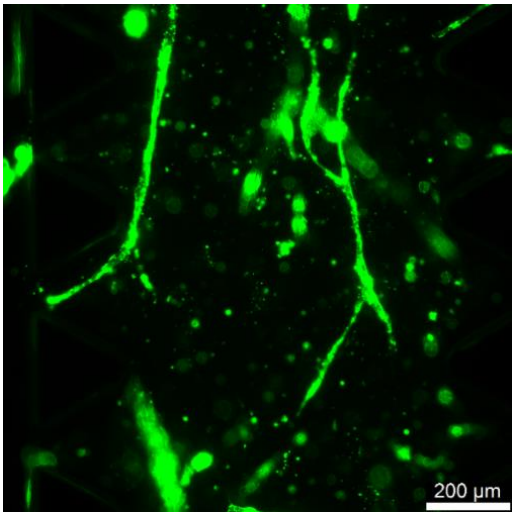


Figure 14. *Neurovascular experiment chip in which mediums were changed only 3 days a week. Photo taken at day 12. Scale bar: 200 µm.*

However, better HUVEC vascularization can be observed in the vascular control chip in Figure 15, than in any type of chip.

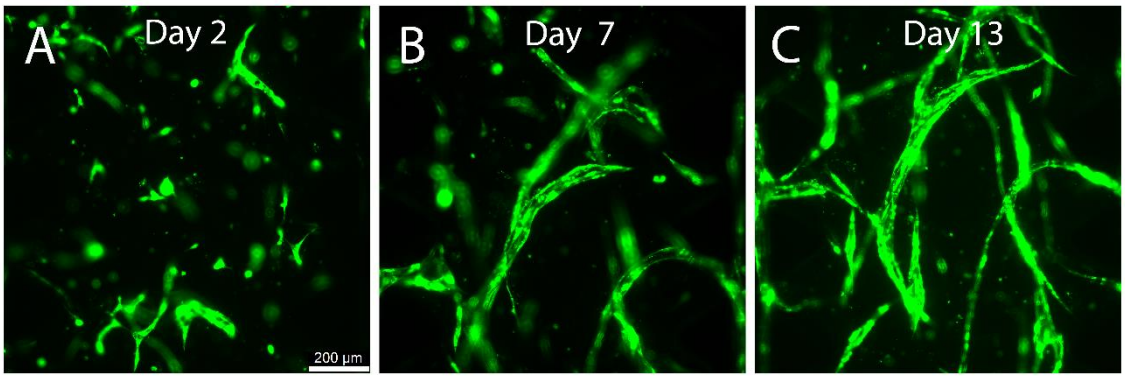


Figure 15. *Vascular control on gravity-driven perfusion. The hydrogel did not degrade. Scale bar: 200 µm.*

In the last figure, Figure 16, shows the vascularization of the previous chip study under hydrostatic pressure perfusion for comparison.

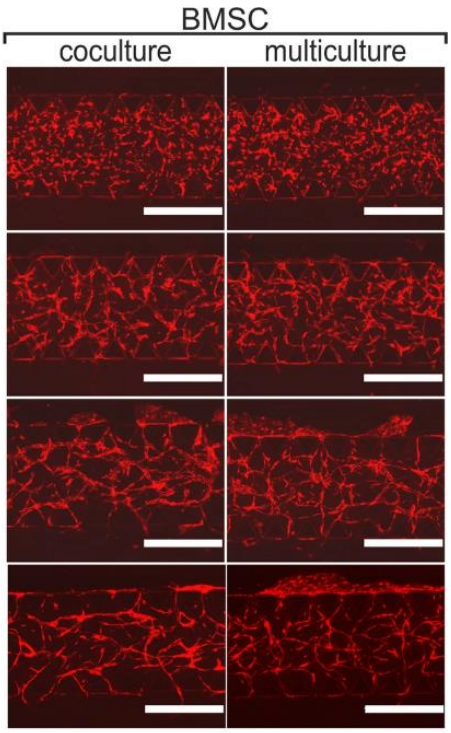


Figure 16. *In comparison, seen in the right column, developed vascularization can be seen as a result of the previous study by Isosaari et al.*

5. DISCUSSION

The aim of study was to experimentally test how the rocker platform induced perfusion primarily affects the formation of neuronal and vascular networks. Perfusion was studied in three different types of cell cultures; neuronal monocultures, vascular cultures, and neurovascular cultures. In addition to rocker platform perfused chips, all cell types were also tested for control by a standard method of perfusion for the AIM Biotech chips, i.e. perfusion caused by gravity-driven hydrostatic pressure.

The outcomes unambiguously indicated that the perfusion method with this context of parameters applied via the rocker platform did not enhance vascularization or the integrity of the hydrogel. The reasons for these observations remain partially unresolved, as this kind of research has not been done much in the literature, in which neurovascular tissue outside the BBB is modeled with a perfusion flow using a rocking plate. NVU-on-chip or BBB-on-chip rocker platform perfusion studies suggest that perfusion promote cell division and better junctional organization indicating that bidirectional perfused media is superior to a static solution. (Wevers et al., 2021) BBB and NVU models often have two different compartments, one with a high flow rate for endothelial cells but slow for neurons which makes a fundamental difference between BBB models and neurovascular tissue models outside of BBB. In this study, rocking platform produces the same shear stress for all cell types, so comparing such 2 compartment studies might be pointless. Furthermore, in studies involving the BBB, perfusions are commonly facilitated through the use of pumps or syringes. (Amirifar et al., 2022)

In general, it is also true in science that fewer studies and images are published if their results do not support the hypothesis, known as “publication bias”, which is why reference material on hydrogels degradation on the neurovascular rocker platform is not readily available. Also, AIM BioTech chips are optimized for hydrostatic pressure perfusion, so they may not be fully suitable for rocker perfusion. In the literature, rocker perfusion has been performed in multiculture/BBB modeling with more experimental chips or with Three-lane OrganoPlate by Mimetas. (Miller and Shuler, 2016; Wevers et al., 2021) Also, Experiments have not been repeated beyond the original study by Isosaari et al. and this experiment, so the discussion is based only on these studies. This must be taken into account.

In this neurovascular perfusion study, the problems observed with vasculature formation can partly be explained by overpopulation of HUVECs on T175 plates where they were

expanded prior to chip plating. Despite the stress experienced by HUVECs, it was evident that one vascular control on gravitational chip showed clear vasculature formation. Thus, variation between replicate samples indicate that samples were not consistently prepared. However, the chip with detectable vasculature formation was an exception, and the no development was detected on rocker plates, mostly because of degradation of the hydrogel.

Imaging was performed with a fluorescence microscope, which was enough already from day 5 onwards to sum up the results, where the hydrogel does not withstand 7° to -7° swing angle of perfusion with a 10-minute tilting time. Hydrogel compositions are base of artificial ECM and for example, neurons need “softer” hydrogel, where the ECs favor more rigid hydrogel. In other perspective, neurons exhibit a preference for interstitial flow characteristics (low flow rate) whereas ECs prefer more CBF like, high flow rate. This remains as one paradox with neurovascular perfusion and partly why 2-compartment models are used in BBB modeling. (Bento et al., 2017; Herzog et al., 2021; Lee et al., 2017)

One debatable variable may be cell density, which would explain at least part of hydrogel degradation. In this sense, it was observed that neuronal monocultures grew more normally in both perfusions and long neurites observed. On the other hand, it is difficult to tell the final development of neurons from phase contrast images, so student Siiri Sihvonen's immunostainings can provide more precise results related to neurons. However, a density-related perspective may indicate that as cell density increases, rocker perfusion becomes less favorable. This finding is supported by the fact that the most undeveloped rocker chips were found in the densest cultures in terms of cell count, i.e. neurovascular cultures, which contained three different cell types. Hydrogels of neuron monoculture were better condition than with vascular or neurovascular cultures. The reduction of the total number of cells in the perfusion chips could be a test of experimental interest.

Another way to explain this phenomenon is that when all cell types are together, they secrete some potentially problematic substance due to upregulated signaling crosstalk on rocker perfusion. Dynamic perfusion accelerates many different signal pathways and mechanotransduction, which also increases the number of metabolites in the culture, and has a strong effect on the EVT phenomenon (Herzog et al., 2021). This protein secretion-assisted degradation is, as in this study, usually circumvented by enzyme inhibitors such as aprotinin. Aprotinin was used in the study, but there is still a possibility that stronger shear stress produces some other degradable compounds.

It is also worth highlighting the intervals between medium refreshments, which were done on the rocker platform only 3 times a week, while the chips perfused by hydrostatic pressure got a change every day. However, this long exchange interval for the medium with rocker platforms did not visibly affect the medium negatively, for example by turning the substance yellow, which would indicate depletion of the medium. To properly study the effects of medium refreshment intervals, it would have been beneficial to include a rocker chip in the experiment for which the medium was changed every day during the 14-day review period. Aprotinin could thus have worked differently.

The use of HUVECs cells in brain tissue modelling or innervated tissues modeling in general has been the subject of frequent debate. The question is, why would we use HUVECs when there is the possibility, like BBB studies, of using endothelial cell types found in the brain, such as human brain microvascular endothelial cells (HBMECs). The problem here with perfusion is mainly that cortical endothelial cells do not respond to perfusion in the same way as HUVECs, which are the gold standard in vascular studies (Medina-Leyte et al., 2020). Some studies have found that HBMECs resist elongation in response to shear stress (Reinitz et al., 2015; Ye et al., 2014). In this thesis, HUVECs were selected based on their availability and use in various vascularized models, including the previously published neurovascular model which was utilized in the experiment (Isosaari et al., 2023). However, it would be interesting to see HBMECs and HUVECs as controls in future perfusion experiments.

Finally, one thing to consider for future experiments is also the possible progressivity of the perfusion flow rate. The degradation of hydrogels occurred already on day 5, so slower perfusion could be used in the beginning. For example, the following experimental setup could be utilized to investigate the effects of flow progressivity – use hydrostatic pressure perfusion for the first 5 days and then turn on rocker perfusion. One could imagine that in the beginning, before the vasculature has developed to receive a larger perfusion fluid, a slower interstitial flow can be considered with the gravity gradient alone.

6. CONCLUSIONS

To summarize, it is clear that more research is needed to draw detailed conclusions. In general, the rocker platform or other dynamic perfusion mechanism has been seen in the literature to aid cell cultures due to continuous flow of fluid. However, in this experiment, the hypothesis was rejected, which used a 10-minute tilt time and a 7-degree angle on a rocking platform. Since the neuron monocultures clearly survived best in perfusion, it can be assumed that the crosstalk interaction caused by vascular cells in perfusion plays a role in the problems manifested in hydrogels. Understanding the science behind perfusion optimization in neurovascular tissue, but outside the blood-brain barrier, is virtually unknown and requires much *in vitro* experimentation.

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